

URYUPINO, Russian Federation

WMO: 307810

Lat: 52.750N

Lon: 120.033E

Elev: 1214

StdP: 14.06

Time Zone: 9.00 (E09)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.5	-39.3	-46.4	0.4	-42.2	-42.8	0.5	-38.8	7.4	0.4	5.3	-3.0	0.2	270	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.9	86.7	67.1	82.8	65.7	79.2	64.2	70.6	80.3	68.5	77.7	66.7	75.5	2.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	105.5	75.3	65.5	98.6	72.5	63.8	92.5	70.4	35.4	79.8	33.6	77.8	32.1	75.6	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	8.7	7.1	-47.0	93.5	4.8	3.4	-50.4	96.0	-53.2	97.9	-55.9	99.8	-59.4	102.3		
(5)			-47.2	75.1	4.5	3.5	-50.4	77.6	-53.0	79.7	-55.5	81.6	-58.7	84.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.0	-18.9	-7.4	11.4	32.5	47.8	60.4	65.5	61.2	47.3	27.8	0.6	-18.3		
	DBStd	31.45	8.00	9.57	12.29	8.27	6.77	6.26	5.13	6.27	7.71	9.78	13.28	9.09		
	HDD50	10027	2137	1607	1197	526	123	5	0	5	141	688	1482	2116		
	HDD65	14362	2602	2027	1662	975	534	163	56	145	532	1153	1932	2581		
	CDD50	1266	0	0	0	1	55	316	481	354	59	0	0	0		
	CDD65	126	0	0	0	0	1	24	72	29	0	0	0	0		
	CDH74	2200	0	0	0	1	95	716	914	449	25	0	0	0		
	CDH80	651	0	0	0	0	18	224	289	119	1	0	0	0		
	WSAvg	1.7	0.5	0.7	2.1	2.9	3.2	2.1	1.6	1.6	2.0	1.9	1.1	0.6		
	PrecAvg	16.9	0.2	0.2	0.1	0.7	1.5	2.6	4.3	3.9	2.1	0.6	0.4	0.3		
(15) Precipitation	PrecMax	22.7	0.6	1.0	0.6	2.1	3.6	7.1	8.4	8.6	4.4	2.6	0.7	1.1		
	PrecMin	12.1	0.0	0.0	0.0	0.0	0.1	0.7	1.0	1.5	0.4	0.0	0.1	0.0		
	PrecStd	3.1	0.1	0.2	0.2	0.6	0.9	1.4	1.8	1.7	1.0	0.6	0.2	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	32.0	50.9	69.4	81.9	91.4	91.3	88.3	77.4	61.7	36.7	13.5		
		MCWB	6.8	24.9	37.1	48.4	57.7	66.4	68.4	68.8	60.5	46.7	30.5	11.7		
	2%	DB	4.1	24.6	44.3	61.6	76.3	86.3	86.9	83.4	72.2	54.8	31.0	5.6		
		MCWB	2.6	20.0	33.7	43.9	54.1	65.4	68.3	67.7	57.7	42.2	26.7	4.5		
	5%	DB	-0.6	17.1	38.5	55.8	71.0	81.6	83.0	78.9	67.6	50.1	26.4	0.7		
		MCWB	-1.6	14.0	30.1	40.5	51.9	63.3	67.3	65.6	56.0	39.5	23.4	-0.1		
	10%	DB	-5.5	11.4	33.6	50.6	66.2	77.8	79.2	74.9	63.4	45.0	21.7	-3.6		
		MCWB	-6.2	9.1	27.2	37.6	49.8	62.1	66.5	63.9	53.4	37.0	19.1	-4.3		
	0.4%	WB	7.1	25.1	37.8	48.6	60.0	71.8	73.5	72.4	64.2	47.2	31.5	12.5		
		MCDB	8.9	31.5	48.8	66.7	74.8	83.9	83.7	82.3	72.0	60.1	35.2	13.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	2.6	19.9	33.9	44.3	56.3	67.9	71.4	69.4	60.7	43.2	26.7	4.5		
		MCDB	4.1	24.5	44.3	59.5	71.3	79.5	81.2	78.9	67.9	52.8	30.2	5.4		
	5%	WB	-1.6	14.0	30.6	41.3	53.5	65.7	69.4	67.4	58.1	40.3	23.3	-0.1		
		MCDB	-0.6	16.9	37.9	53.8	68.3	77.7	78.2	76.1	64.3	48.8	26.5	0.7		
	10%	WB	-6.2	9.1	27.4	38.4	51.0	63.8	67.6	65.3	55.3	37.3	19.0	-4.3		
		MCDB	-5.5	11.2	33.4	49.0	64.1	75.2	76.0	72.4	61.9	45.3	21.5	-3.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	26.5	35.8	36.5	30.2	31.2	31.4	25.9	25.1	28.4	28.1	26.8	24.1		
		MCDBR	28.3	39.3	37.4	40.7	42.8	40.1	34.9	33.4	35.4	34.4	28.1	25.2		
		MCWBR	26.7	34.4	27.0	23.8	22.2	20.1	17.4	18.1	21.4	21.9	23.4	24.0		
		MCDBR	28.1	38.6	35.4	36.1	36.4	33.5	28.2	27.6	28.0	33.5	27.4	25.2		
	5% WB	MCWBR	26.5	33.9	25.7	21.8	19.7	18.3	15.8	16.3	18.4	21.6	23.0	24.0		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.290	0.340	0.463	0.535	0.500	0.452	0.423	0.369	0.333	0.290	0.271		
	taud		2.152	2.118	2.054	1.892	1.801	2.005	2.212	2.269	2.355	2.325	2.214	2.085		
	Ebn at Noon		228	256	264	241	230	241	252	253	255	239	213	193		
	Edh at Noon		22	31	41	55	64	53	42	38	30	25	21	19		
	RadAvg		429	788	1302	1706	1705	1871	1688	1474	1130	758	469	325		
(45) All-Sky Solar Radiation	RadStd		12	19	48	96	130	154	141	128	93	52	18	13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page